Rohit Menon

Robotics Researcher | Active Perception, Reasoning and Manipulation

Bonn, Germany | rohit.menon@uni-bonn.de

Website | Google Scholar (242 citations; h-index: 9; i10-index: 9) | GitHub | LinkedIn

Citation metrics as of October 2026

Research Profile

- PhD candidate developing **prior-guided active perception** for robot mapping, object search and manipulation.
- Former **founding engineer and AI group leader at Neura Robotics**, with experience in robot perception, manipulation and real-world deployment.

Education

PhD in Computer Science Feb. 2022–Mar. 2027 (Expected)
University of Bonn, Germany | Advisor: Prof. Dr. Maren Bennewitz

MSc in Automation and Robotics Oct. 2012–Mar. 2015
TU Dortmund University, Germany | Grade: 1.2; top 5% of cohort

Bachelor's in Electrical Engineering Aug. 2004–June 2008
SPCE, University of Mumbai, India | Grade: 76%; silver medal

Research Experience

Research Associate / PhD Researcher Feb. 2022–Present
Humanoid Robots Lab, University of Bonn, Germany

- Authored **six first- or equal-first-author peer-reviewed papers** on active perception, manipulation, scene graphs and uncertainty-aware mapping.
- Coauthored **twelve additional peer-reviewed papers** in robot perception, manipulation, semantic mapping and human–robot interaction.
- Supervised **five completed and two ongoing MSc theses, five lab projects and 10+ seminar projects**; mentored **three junior PhD students**.

  PhenoRob
  - Developed shape-completion-guided next-best-view planning for fruit mapping, validated on a UR5e in a commercial greenhouse (NBV-SC).
  - Developed a three-arm pepper harvesting system using active perception, online scene graphs and collision-aware IK (HortiBot; DawnIK).
  - Developed uncertainty-aware depth and semantics for 3D mapping; evaluated on NYUv2 and zero-shot ScanNet (EvidMTL/EvidKimera).

  Lamarr
  - Developed open-vocabulary object search using LLM location priors, scene geometry and dynamic probabilistic scene graphs (O-STaR).
  - Developed semantic reasoning for search in dynamic and concealed spaces; received the IROS 2025 workshop Best Poster Award (paper).

  AID4Crops
  - Integrated uncertainty-aware depth completion, persistent panoptic mapping and plant scene graphs with view-motion planning (SG-AMP).
  - Developed graph reasoning to infer plant-organ relations and rank leaf occlusions for robotic fruit intervention (SG-DOR).

Researcher, Robot Manipulation and Control July 2015–Oct. 2019
DFKI Robotics Innovation Centre, Bremen, Germany

- Developed **planning, control, teleoperation and manipulation software** for high-DoF, mobile and humanoid robots.
- Led DFKI's iLEVATOR work, including **technical coordination, budget, procurement and reporting**; supervised one MSc thesis.

  NeTTUN
  – Developed the control and communication architecture for a tunnel-boring manipulator, including jerk-limited trajectories and teleoperation.

  iLAADR / iLEVATOR
  – Built monocular known-object picking from CAD templates; iLEVATOR also placed depleted bins on a conveyor at the end-user factory.

  Hybr-iT
  – Developed mobile-manipulation planning and trajectory generation, integrated on MobiPick for Hannover Messe 2019.

  TransFit
  – Built unknown-object picking with learned grasping and optimization-based planning; developed tactile/reactive grasping and RH5 arm planning.

MSc Thesis Researcher, Shared Autonomy for Assistive Robotics June 2014–Feb. 2015
DLR Institute of Robotics and Mechatronics, Oberpfaffenhofen, Germany

- Developed **adaptive shared autonomy** for assistive manipulation using automatic grasp selection and **virtual-fixture guidance**.
- Preserved continuous user control of the object approach direction while guiding velocity commands toward a selected grasp pose.
- Validated the method in **full-dynamics simulation, a user study and on the real robot**; thesis grade: 1.3; published at ICRA 2016.

MSc Research Project, Neuro-Evolutionary Robot Navigation During MSc, 2012–2015
TU Dortmund University, Germany

- Investigated **neuro-evolutionary learning of echo state networks** for **mobile robot navigation**.

Teaching and Supervision

MSc Theses

- Supervised **five MSc theses** at the University of Bonn: **two conference papers, one workshop paper and two thesis awards**.
- Supervised research on **active perception, fruit mapping and manipulation**.
- At DFKI, supervised an MSc thesis on **multi-sensor fusion for in-hand object localization**, slip compensation and placement.

PhD Mentoring

- Supervising **three junior PhD students** on interactive perception and manipulation at the Humanoid Robots Lab.

MSc Lab Projects

- **Summer 2026:** Open-Vocabulary Mobile Manipulation – **Active Perception**; Open-Vocabulary Mobile Manipulation – **Flexible Command Grounding**.
- **Winter 2025/26: Robot Butler**: Open-Vocabulary Mobile Manipulation; **Open Sesame!**: Whole-Body Control for Opening Furniture.
- **Winter 2024/25: Rock–Paper–Scissors** (co-supervised).

Seminars

- Supervised 10+ **Humanoid Robots seminar projects** (2022–2026) on active perception, manipulation and scene understanding.

Humanoid Robotics Lecture **Summer 2025, 2026**

- Redesigned **manipulation and perception lectures**, exercises and assessments; integrated modern perception pipelines and semantic mapping.
- Conducted oral and written examinations and managed grading.

Cognitive Robotics Lecture **2022, 2023, 2024**

- Served as **stand-in lecturer and teaching assistant** for probabilistic robotics; designed assignments and examinations.
- Supervised practical programming assignments and managed grading.

Earlier Teaching

- Graded **Reinforcement Learning** tutorials at the University of Bremen during the DFKI appointment.
- Organized a one-day **MATLAB workshop** for undergraduate engineering students at SPCE in 2008.

Academic Service

- **Journal reviewer:** IEEE Robotics and Automation Letters (RA-L); The International Journal of Robotics Research (IJRR).
- **Conference reviewer:** ICRA, IROS, CASE, Humanoids; Robotics: Science and Systems (RSS), 2026.
- Served on the **PhD interview committee** at the Humanoid Robots Lab, University of Bonn.
- Member of the **Robotics Institute Germany technical cluster on semantic perception**.
- Organizing the **PhenoRob Study Group on Photosynthesis**.

Research Funding and Project Development

- Helped write the **Agriloop proposal (EUR 400,000)** on foundation models for human-in-the-loop greenhouse robotics.
- Contributed to the **PhenoRob Phase 2 HRL PhD proposal**.

Invited Talks

Active Perception for Robotics in Agriculture May 2025
Pint of Science Festival, Bonn, Germany; public-facing talk on active perception for crop monitoring and harvesting.

Active Perception Dec. 2022
Indian Institute of Information Technology (IIIT) Kottayam, India.

Outreach and Public Engagement

- Featured in a **WDR Bonn television interview**, demonstrating HortiBot and discussing robotic horticulture.
- Demonstrated **HortiBot** during **NRW State Premier Hendrik Wüst's visit** to the University of Bonn, February 2024.
- Presented **active perception and fruit mapping** at the ICRA@40 live demonstration, Rotterdam, 2024.
- Gave robot demonstrations at **Tag der offenen Tür (2024, 2025)**, **Wissenschaftsfest (2024)** and **Bonn Nacht der Technik (2023)**.
- Demonstrated Neura Robotics robots at **GITEX 2021**, Dubai, with coverage in *The National*.

Industry Experience

Founding Engineer / Group Leader, AI for Manipulation Nov. 2019–Jan. 2022
Neura Robotics GmbH, Metzingen, Germany

- **Built MAiRA's perception–manipulation architecture** as the first AI engineer, including C++/ROS libraries, drivers and interfaces.
- Rewrote **Xenomai/Orocos RTT tasks for hard real-time control and actuator communication** via SOEM EtherCAT.
- Developed selective pick-and-place using semantic segmentation and depth fusion; integrated **voice- and gesture-based manipulation commands**.
- Built **human collision avoidance using RGB, depth and radar fusion** with 3D pose tracking; developed a tic-tac-toe demonstrator.
- Promoted to **AI for Manipulation group leader** in September 2021, with **hiring responsibility**.

Engineering and Commissioning / Senior Executive Sept. 2008–Sept. 2012
Siemens Ltd, Thane, India

- Developed **AC/DC drive controls and PLC/HMI software**; commissioned machinery, tuned control loops and optimized machine logic.
- Managed technical interfaces with clients, consultants and vendors; progressed from graduate trainee to executive and senior executive.

Awards, Scholarships and Honours

Best Poster Award 2025
IROS Workshop on Perception and Planning for Mobile Manipulation in Changing Environments.

Deutschlandstipendium 2013
Scholarship for academic excellence, TU Dortmund University.

Siemens Excellence in Commissioning Trophy 2011
Siemens India Conference.

Silver Medal, Bachelor's in Electrical Engineering 2008
Sardar Patel College of Engineering, University of Mumbai.

Ratan Tata Scholarship 2006, 2007
Scholarship for excellence in engineering during the bachelor's degree.

Best Student in Science Faculty 2004
Bhavan's Junior College, Mumbai; all-round performance among more than 1,000 students.

First Rank, Class XII HSC Examination 2004
Bhavan's Junior College, Mumbai.

Second Rank, Class X SSC Examination 2002
IES Secondary School, Mumbai.

Research Methods and Technical Skills

Methods: Active perception; scene graphs; graph neural networks; shape completion; evidential deep learning; uncertainty quantification; semantic mapping; open-vocabulary mobile manipulation; shared autonomy; neuro-evolution and echo state networks.
Programming and learning: C++, Python, MATLAB/Simulink, PyTorch, PyTorch Geometric.
Robotics and simulation: ROS, ROS 2, Isaac Sim, Gazebo, Blender.
Control and libraries: Xenomai, Orocos RTT, SOEM EtherCAT, OpenCV, PCL, Eigen, NLopt.
Languages: English C2 (TOEFL: 116/120), German B2, Hindi, Malayalam.

Publications

Full author lists; **Rohit Menon** is highlighted; * denotes equal first-author contribution. Published versions and their arXiv preprints are listed once.

Journal Articles

[1] Rajasree Padmakumari Hemachandran Nair, **Rohit Menon**, Ralf Kemkemer. Generalised Image Processing Method for Quantitative Analysis of Nucleus, Cell and Focal Adhesion Clusters. *Current Directions in Biomedical Engineering*, 7(2), pp. 558–561, 2021.

[2] Srinidhi Balasubramanian, Donald Michael McFarland, Sotiria Koloutsou-Vakakis, Kan Fu, **Rohit Menon**, Christopher Lehmann, Mark J. Rood. Effect of Grid Resolution and Spatial Representation of NH_3 Emissions from Fertilizer Application on Predictions of NH_3 and $PM_{2.5}$ Concentrations in the United States Corn Belt. *Environmental Research Communications*, 2(2), article 025001, 2020.

Conference Publications

[3] **Rohit Menon***, Niklas Mueller-Goldingen*, Sicong Pan, Gokul Krishna Chenchani, Maren Bennewitz. SG-DOR: Learning Scene Graphs with Direction-Conditioned Occlusion Reasoning for Pepper Plants. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[4] Benno Wingender*, Nils Dengler*, **Rohit Menon**, Sicong Pan, Maren Bennewitz. A Robust Placeability Metric for Model-Free Unified Pick-and-Place Reasoning. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[5] **Rohit Menon**, Yasmin Schmiede, Gokul Krishna Chenchani, Hermann Blum, Maren Bennewitz. O-STaR: Open-Vocabulary Object Search through Spatio-Temporal Reasoning on Dynamic Scene Graphs. *German Robotics Conference (GRC)*, 2026.

[6] **Rohit Menon**, Nils Dengler, Sicong Pan, Gokul Krishna Chenchani, Maren Bennewitz. EvidMTL: Evidential Multi-Task Learning for Uncertainty-Aware Semantic Surface Mapping from Monocular RGB Images. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[7] Allen Isaac Jose*, Sicong Pan*, Tobias Zaenker, **Rohit Menon**, Sebastian Houben, Maren Bennewitz. GO-VMP: Global Optimization for View Motion Planning in Fruit Mapping. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[8] Ahmed Shokry, Walid Gomaa, Tobias Zaenker, Murad Dawood, **Rohit Menon**, Shady A. Maged, Mohammed I. Awad, Maren Bennewitz. Context-Based Meta Reinforcement Learning for Robust and Adaptable Peg-in-Hole Assembly Tasks. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[9] Nils Dengler, Jesper Mücke, **Rohit Menon**, Maren Bennewitz. Efficient Manipulation-Enhanced Semantic Mapping With Uncertainty-Informed Action Selection. *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2025.

[10] Christian Lenz*, **Rohit Menon***, Michael Schreiber, Melvin Paul Jacob, Sven Behnke, Maren Bennewitz. HortiBot: An Adaptive Multi-Arm System for Robotic Horticulture of Sweet Peppers. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

[11] Benedikt Kreis, Nils Dengler, Jorge de Heuvel, **Rohit Menon**, Hamsa Perur, Maren Bennewitz. Compact Multi-Object Placement Using Adjacency-Aware Reinforcement Learning. *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2024.

[12] Salih Marangoz*, **Rohit Menon***, Nils Dengler, Maren Bennewitz. DawnIK: Decentralized Collision-Aware Inverse Kinematics Solver for Heterogeneous Multi-Arm Systems. *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2023.

[13] **Rohit Menon**, Tobias Zaenker, Nils Dengler, Maren Bennewitz. NBV-SC: Next Best View Planning Based on Shape Completion for Fruit Mapping and Reconstruction. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

[14] Tobias Zaenker, Julius Rückin, **Rohit Menon**, Marija Popović, Maren Bennewitz. Graph-based View Motion Planning for Fruit Detection. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

[15] Nils Dengler, Sicong Pan, Vamsi Kalagaturu, **Rohit Menon**, Murad Dawood, Maren Bennewitz. Viewpoint Push Planning for Mapping of Unknown Confined Spaces. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

[16] Benedikt Kreis, **Rohit Menon**, Bharath Kumar Adinarayan, Jorge de Heuvel, Maren Bennewitz. Reactive Correction of Object Placement Errors for Robotic Arrangement Tasks. *International Conference on Intelligent Autonomous Systems (IAS-18)*, pp. 257–270, 2023.

[17] Salih Marangoz, Tobias Zaenker, **Rohit Menon**, Maren Bennewitz. Fruit Mapping with Shape Completion for Autonomous Crop Monitoring. *IEEE International Conference on Automation Science and Engineering (CASE)*, 2022.

[18] Jörn Vogel, Katharina Hertkorn, **Rohit U. Menon**, Máximo A. Roa. Flexible, Semi-Autonomous Grasping for Assistive Robotics. *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4872–4879, 2016.

Workshop Papers and Posters

[19] Benno Wingender, Nils Dengler, **Rohit Menon**, Sicong Pan, Maren Bennewitz. Robust Placeability Estimation for Model-Free Unified Pick-and-Place Reasoning. *ICRA Workshop on Manipulation Robustness: Towards Human-Level Robustness under Real-World Challenges*, 2026.

[20] **Rohit Menon**, Yasmin Schmiede, Maren Bennewitz, Hermann Blum. Open-Vocabulary and Semantic-Aware Reasoning for Search and Retrieval of Objects in Dynamic and Concealed Spaces. *IROS Workshop on Perception and Planning for Mobile Manipulation in Changing Environments*, 2025. *Best Poster Award.*

[21] Nils Dengler, Sicong Pan, Vamsi Kalagaturu, **Rohit Menon**, Murad Dawood, Maren Bennewitz. Learning Interactive Viewpoint Planning for Foresighted Object Discovery Considering Human Preferences. *ICRA Workshop on Human-Aligned Reinforcement Learning for Autonomous Agents and Robots*, 2024.

[22] José de Gea Fernández, Dennis Mronga, Martin Günther, Sebastian Stock, Nils Niemann, Hendrik Wiese, **Rohit Menon**, Elsa Andrea Kirchner, Stefan Stiene. Towards Contextual Robots for Collaborative Manufacturing. *IROS Workshop on Human-Robot Cooperation and Collaboration in Manipulation: Advancements and Challenges*, 2018. *Poster.*

Preprints and Manuscripts under Review

[23] **Rohit Menon**, Shiva Rudra Lolla, Niklas Mueller-Goldingen, Gokul Chenchani, Ribana Roscher, Maren Bennewitz. SG-AMP: Scene-Graph-Guided Active Perception and Semantics-Aware Motion Planning for Pepper Plants. *arXiv:2609.01579*, 2026. *Preprint.*

[24] **Rohit Menon**, Yasmin Schmiede, Gokul Krishna Chenchani, Hermann Blum, Maren Bennewitz. O-STaR: Effective Object Search through Spatio-Temporal Reasoning on Dynamic Scene Graphs. *Manuscript*, 2026. *Extended version submitted to ICRA 2027; under review.*

[25] Sicong Pan, Luca Lobefaro, Moein Taherkhani, Xuying Huang, **Rohit Menon**, Cyrill Stachniss, Maren Bennewitz. Efficient View Planning Guided by Previous-Session Reconstruction for Repeated Plant Monitoring. *arXiv:2510.07028*, 2025. *Preprint; revised in 2026; submitted for review.*